

Objective

The main objective of this task is to develop an Employee Salary Calculator that calculates an employee's gross salary, deductions, and final net salary based on predefined and configurable salary rules. This task helps in practicing Python dictionaries, functions, conditional validation, arithmetic operations, and implementing business rules in a practical application. The Streamlit interface makes the calculator interactive and easy to use.

Key Objectives

- Calculate employee gross salary.
- Calculate different salary allowances.
- Calculate deductions such as PF and tax.
- Calculate final net salary.
- Practice Python functions and dictionaries.
- Implement configurable salary rules.
- Perform input validation.
- Build an interactive Streamlit application.

3. Key Features

The application allows the user to enter an employee name and basic salary and then automatically calculates the complete salary breakdown. It calculates HRA, travel allowance, gross salary, PF deduction, tax deduction, total deductions, and final net salary. The application also validates the employee name and salary input to prevent invalid calculations. Salary rules are maintained separately in the program, making them easy to modify when requirements change.

Main Features

- Employee name input
- Basic salary input
- HRA calculation
- Travel allowance calculation
- Gross salary calculation

- PF deduction
- Tax deduction
- Total deductions
- Net salary calculation
- Input validation
- Salary breakdown
- Configurable salary rules
- Streamlit local web interface

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY16>

```

1 import streamlit as st
2 st.set_page_config(
3     page_title="Employee Salary Calculator",
4     page_icon="👤",
5     layout="centered"
6 )
7
8 # Title
9 st.title("👤 Employee Salary Calculator")
10 st.write("Calculate gross salary, deductions, and net salary based on configurable salary rules.")
11
12 st.divider()
13
14 # Salary rules
15 BASIC_PERCENTAGE = 50
16 HRA_PERCENTAGE = 20
17 TA_PERCENTAGE = 10
18 PF_PERCENTAGE = 12
19 TAX_PERCENTAGE = 5
20
21 def calculate_salary(basic_salary):
22     hra = basic_salary * HRA_PERCENTAGE / 100
23     ta = basic_salary * TA_PERCENTAGE / 100
24
25     gross_salary = basic_salary + hra + ta
26
27     pf = basic_salary * PF_PERCENTAGE / 100
28     tax = gross_salary * TAX_PERCENTAGE / 100
29
30     total_deductions = pf + tax
31     net_salary = gross_salary - total_deductions
32
33     return {
34         "Basic Salary": basic_salary,
35         "HRA": hra,
36         "Travel Allowance": ta,
37         "Gross Salary": gross_salary,
38         "PF": pf,
39         "Tax": tax,
40         "Total Deductions": total_deductions,
41         "Net Salary": net_salary
42     }
43
44 st.subheader("👤 Employee Details")
45
46 employee_name = st.text_input("Employee Name")
47
48 basic_salary = st.number_input(

```

```

48     "Basic Salary (₹)",
49     min_value=0.0,
50     step=1000.0,
51     format="%.2f"
52 )
53
54 # Calculate button
55 if st.button("Calculate Salary", type="primary"):
56
57     if not employee_name.strip():
58         st.warning("Please enter the employee name.")
59
60     elif basic_salary <= 0:
61         st.warning("Please enter a valid basic salary greater than 0.")
62
63     else:
64         result = calculate_salary(basic_salary)
65
66         st.success(f"Salary calculated successfully for {employee_name}!")
67
68         st.divider()
69
70         # Salary breakdown
71         st.subheader("💰 Salary Breakdown")
72
73         col1, col2 = st.columns(2)
74
75         with col1:
76             st.metric("Basic Salary", f"₹{result['Basic Salary']:, .2f}")
77             st.metric("HRA", f"₹{result['HRA']:, .2f}")
78             st.metric("Travel Allowance", f"₹{result['Travel Allowance']:, .2f}")
79             st.metric("Gross Salary", f"₹{result['Gross Salary']:, .2f}")
80
81         with col2:
82             st.metric("PF Deduction", f"₹{result['PF']:, .2f}")
83             st.metric("Tax Deduction", f"₹{result['Tax']:, .2f}")
84             st.metric("Total Deductions", f"₹{result['Total Deductions']:, .2f}")
85             st.metric("Net Salary", f"₹{result['Net Salary']:, .2f}")
86
87         st.divider()
88
89         st.subheader("🏆 Final Salary")
90
91         st.success(
92             f"Net Salary of {employee_name}: "
93             f"₹{result['Net Salary']:, .2f}"
94         )

```

OUTPUT: <http://localhost:8501/>



Employee Salary Calculator

Calculate gross salary, deductions, and net salary based on configurable salary rules.



Employee Details

Employee Name

Sharvari

Basic Salary (₹)

49750.00

- +

Calculate Salary

Salary calculated successfully for Sharvari!



Salary Breakdown

Basic Salary

₹49,750.00

PF Deduction

₹5,970.00

HRA

₹9,950.00

Tax Deduction

₹3,233.75

Travel Allowance

₹4,975.00

Total Deductions

₹9,203.75

Gross Salary

₹64,675.00

Net Salary

₹55,471.25



Final Salary

Net Salary of Sharvari: ₹55,471.25

View Salary Rules

HRA: 20% of Basic Salary

Travel Allowance: 10% of Basic Salary

PF: 12% of Basic Salary

Tax: 5% of Gross Salary

Linkedin:-<https://lnkd.in/p/dnWH-3TN>



Employee Salary Calculator

Calculate gross salary, deductions, and net salary based on configurable salary rules.

Employee Details

Employee Name

Sharvari

Basic Salary (₹)

49750.00

Calculate Salary

Salary calculated successfully for Sharvari!



Shambhavi E L • You
Student at The Oxford College of ...
Just now

VEDA Internship – Python Programming Track | Day 16 Excited to complete Day 16 of my Python Programming Internship journey! 🎉 Today, I built an Employee Salary Calculator using Python and Streamlit. The application calculates gross salary... more

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Github:-<https://github.com/shambhavi0804/VEDA-internship/commit/0b1a8068ff8b3246e129c03b7bd9418346410dba>

```
Filter files...
DAY16
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DAY16/app.py

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```